# Do Now · retrieval

## Waterfall, Agile and Extreme Programming

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Which OCR pseudocode function reads the next line of a text file?

### Q2

What is a function parameter?

### Q3

Why use boundary-value testing?

### Q4

If requirements frequently change, what is the risk of freezing all requirements before implementation?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Waterfall is sequential — Agree requirements, then design, implementation, testing and deployment. Revisiting an earlier stage can be costly.

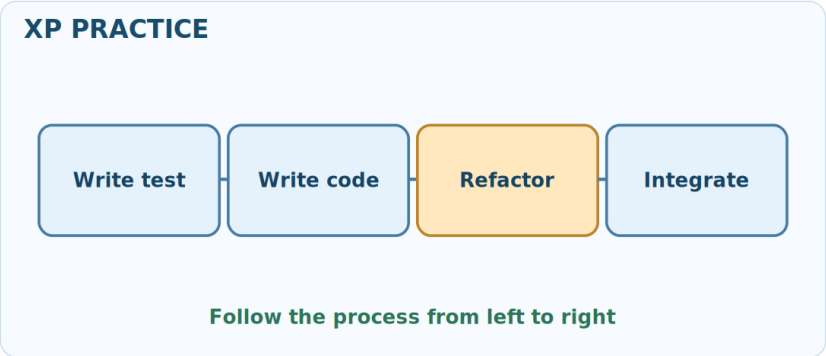
## Core idea 2

Agile delivers incrementally — Each short iteration produces working software and seeks feedback, so priorities can change between iterations.

## Core idea 3

Extreme Programming (XP) — XP is an agile methodology emphasising practices such as test-driven development, pair programming, refactoring and continuous integration.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

Waterfall — Primarily sequential software-development life cycle.

Agile — Iterative approach using feedback and incremental releases.

XP — Extreme Programming, an agile methodology with disciplined engineering practices.

sprint — Short time-boxed development iteration.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

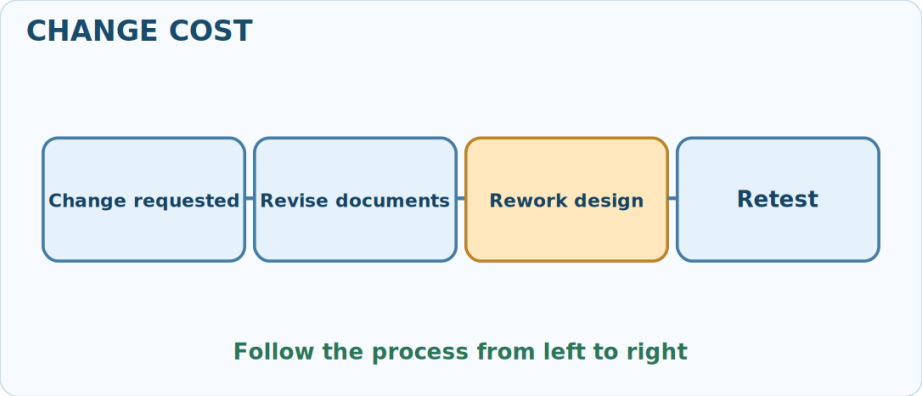
A hospital requests software with a carefully agreed specification and formal approvals. Why might Waterfall suit it?

## Follow the worked method

1. Predict first — Requirements and approval milestones can be documented before later stages proceed.

2. Model and explain — Waterfall offers clear documentation and control, but late changes can require expensive rework.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first

Changing needs and quick feedback favour Agile iterations over a rigid up-front plan.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

State a defining characteristic of Waterfall, Agile and XP.

## Question 2

A mobile app company receives new customer requirements each fortnight. Recommend a methodology and explain two advantages and one disadvantage.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain how test-driven development and continuous integration in XP could reduce defects, and why neither guarantees fault-free software.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

Agile means software is never planned or tested.

## Use key vocabulary from memory

1. Explain Waterfall in your own words.

2. Explain Agile in your own words.

3. Explain XP in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.